

General introduction

Empirical and Theoretical Research Methodology

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Learning goals and structure

Introduction to theoretical and empirical research methodology

Lecture

- Weekly in-person meeting
- Introduces key concepts, tools, and methods
 - Literature reviews
 - Interviews
 - AI Tools
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- Foundations for upcoming seminars

Implementation lab

- Alternating between face-to-face event and instructional videos
- Programming language R
- Focus: practical implementation of key tools and methods
- Additional topics such as visualization and data prep

Tentative schedule

For the most recent version,
see the course webpage!

Lecture

1. General introduction
2. AI tools in practice
3. Basics of logic
4. Logic, argumentation, models, and explanation
Take Home Examination
5. Quantitative and qualitative research
6. Interviews
7. Case studies
8. Literature reviews
Take Home Examination
9. Statistics (recap)
10. Regression analysis
Take Home Examination
11. Experiments
12. Overall questions

Implementation lab

1. General introduction and installation (👩👩)
2. Basics of R and R-Studio (🎥🎞)
3. Project Management (👩👩)
4. Object types (🎥🎞)
5. Quarto and R Markdown (🎥🎞)
6. Importing data (🎥🎞)
7. Preparing data (👩👩)
8. Data visualisation (👩👩)
9. Using AI for coding (👩👩)
10. Linear models (🎥🎞)
11. Analysing experimental data (🎥🎞)
12. Recap & practice (👩👩)

Goal: learn essential tools for management research

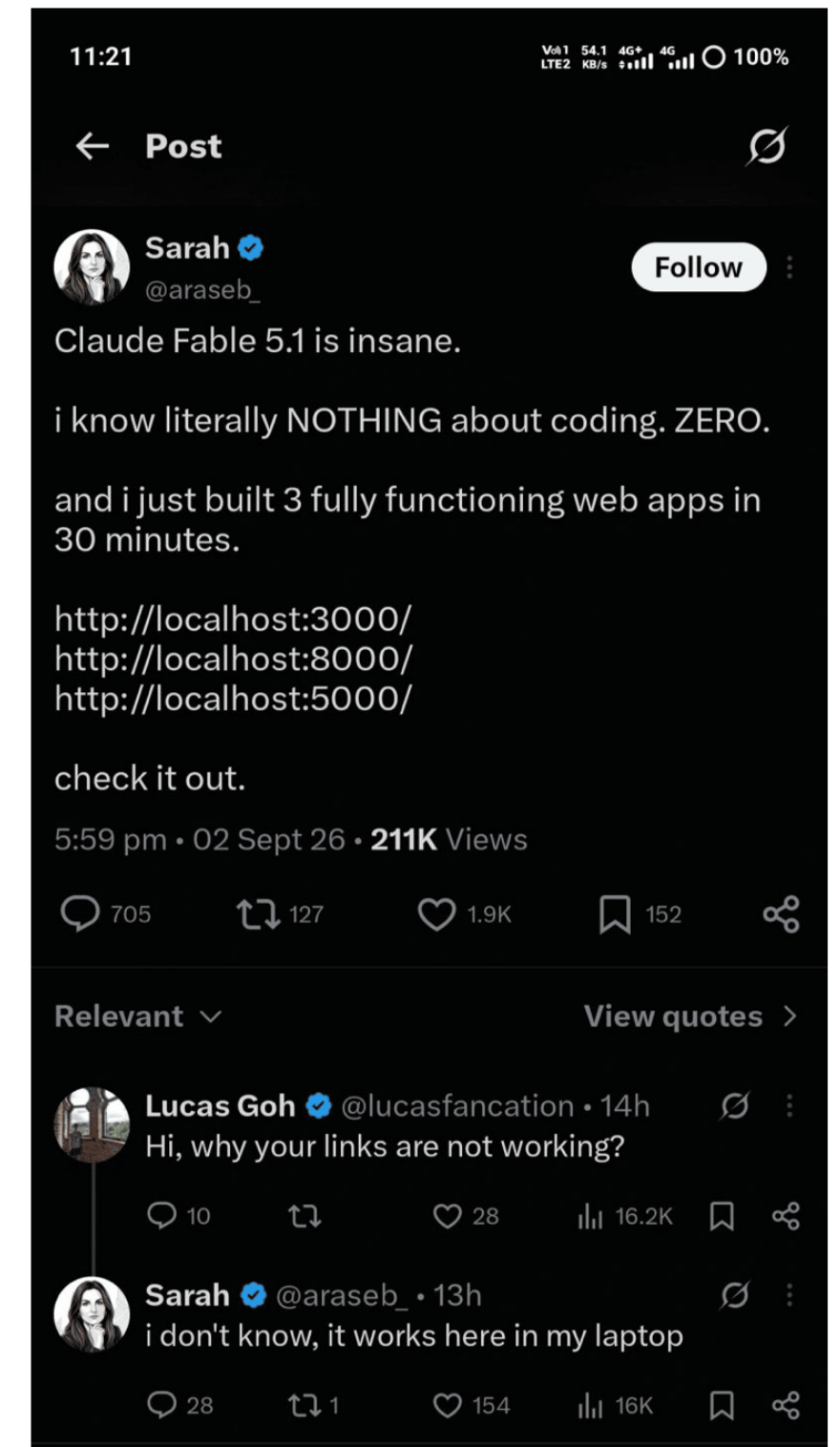
- Practical exercises, especially for the R content
 - Work together, find study groups
- Ask questions and provide feedback
 - Use the Moodle forum!
- Follow both parts of the module constantly
 - Do the exercises provided
- Work constantly; the module is rather time-intensive
 - Honest 5 ECTS: 150 hours work

Artificial Intelligence in this module

- Teaching AI tools is one learning goal of this module (but not a central one)
- AI has become very relevant in practice → we are not against using it
- **Experience:** AI makes good students better and lazy students worse
- You are allowed to use any AI tool you want
 - This includes the final exam and take-homes → but **only free plans**
 - You are required to **document** your use of AI completely
 - Undocumented use can be considered **punishable fraud**
- For your own sake: use AI in a **self-reflective way**, and do not rely on it too much

AI and coding

- This module teaches coding - isn't this outdated?
- No: coding is much more than typing code
 - Even professionals write less and less code themselves
 - They now think, architect, plan, prompt, test, revise...
 - The AI does the typing
- Typing code is gone; coding more relevant than ever
- You can leverage AI only if you know coding as a whole



Logistics

- Upon successful completion, this course is worth **5 CP**
 - Corresponds to **150 working hours**, about 25 being lecture time
- Discussion, feedback forms and announcements via **Moodle**
 - Moodle room: [17920](#) | Moodle password: ResearchMethods26
 - Most important: the **forum** for our questions and announcements
- The material will be made available via a [course webpage](#)
 - Written in R → easier to maintain + makes material publicly available

Examination

- **Prerequisite** for final exam: pass 2 out of 3 take-home examinations
 - See course outline for the timing of the take homes
 - You must register for the take-homes (Prüfungsvorleistung)
 - Take-homes are digital and completed at home, but on your own
 - If you fail two take-home examinations, **you cannot complete the course**
- **Content of the final exam (completely digital)**
 - 1/3 - 1/2: Multiple choice questions in Moodle
 - 1/3 - 1/2: Open questions and technical applications
 - 1/3 - 1/2: R tasks
- **Open book character** is meant to mimic the practical application of the tools

Prerequisites

- For most topics, no specific prior knowledge is required
- We must assume **foundational knowledge in mathematics and statistics**
 - All material from the "**A Recap of Undergraduate Statistics**" Track
 - <https://euf-methodology-hub.netlify.app/tracks/statistics-recap>
 - Will be examined in third take-home examination
 - You have difficulties with some topics → start addressing them **now**
- Ask your questions for the recap session until November 16!

Summary: our 'learning agreement'

The goal

You learn scientific methods and tools that are essential to participate in advanced seminars at the University and that are relevant in advanced management activities. You learn to use modern AI tools and the programming language R to conduct applied research tasks.

What we offer

We provide **slides, instructional videos, example codes, tutorials, and exercises** that are tailored to your learning needs. We will give our best to facilitate an **amicable working environment** and answer questions in class and via Moodle. We seek your **feedback** and implement it, when feasible.

What we expect

We expect you to **attend** classes regularly, to be **honest** about what you did not understand, to **support each other** through Moodle and in class, that you do the **coursework** and **exercises regularly** such that you keep up with the course, and that you give **feedback**.

Some (personal) software recommendations I

Generative AI Assistant

- Free plan quite good
- Constitutional AI
- Very good writing capabilities



For literature management

- Free and open source
- Cross-platform, many features
- Compatible with many other apps
- Requires fixed investment, but pays off



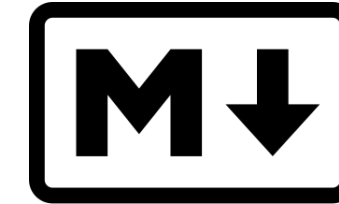
For note-taking

- Completely free and flexible
- Cross-platform, many features
- Blog: <https://obsidian.rocks/>



Some (personal) software recommendations II

For writing



- Markdown: widespread and simple markup language for creating formatted text
- Free and app-independent
- Quarto: extension to integrate programming languages
- Helps creating docx, pdf, html,...
- Very easy to learn by examples



For working with data

- Widely used in Academia and Business
- Free and open source
- Great for preparing, analysing and visualising quantitative data



For translating and editing

- Translation and style improvement with DeepL
- Grammarly for spelling and grammar, good app integration
- Free plans are already useful



DeepL



Open Questions?
